Check for updates

ACM DIGITAL LIBRARY

Association for Computing Machinery

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3530019.3531331

RESEARCH-ARTICLE

The Influence of Cost Drivers on Effort Estimation in Distributed Software Development

DANISH IQBAL, Masaryk University, Brno, Czech Republic

SHER BADSHAH, Dalhousie University, Halifax, NS, Canada

IRFAN KAZIM, COMSATS University Islamabad, Islamabad, Pakistan

Open Access Support provided by:

Dalhousie University

Masaryk University

COMSATS University Islamabad

PDF Download
3530019.3531331.pdf
26 January 2026
Total Citations: 0
Total Downloads: 147

Published: 13 June 2022

Citation in BibTeX format

EASE 2022: The International Conference on Evaluation and Assessment in Software Engineering 2022
June 13 - 15, 2022
Gothenburg, Sweden

EASE '22: Proceedings of the 26th International Conference on Evaluation and Assessment in Software Engineering (June 2022)
https://doi.org/10.1145/3530019.3531331
ISBN: 9781450396134

The Influence of Cost Drivers on Effort Estimation in Distributed Software Development

Danish Iqbal, Ijaz ul haq and Irfan Kazim
Faculty of Informatics, Masaryk University, Brno Czech Republic
Faculty of Education, Psychology and Social Work, University of Lleida, 25003 Lleida, Spain
Department of Computer science COMSATS University Islamabad, Islamabad, Pakistan
danish@mail.muni.cz, ijaz.ul-haq@udl.cat ,erfanmumtaz@gmail.com

ABSTRACT

Nowadays, software projects are a vital element in any organization's success. It plays the highest role in the organization's success in most cases. Hence, its main focus is to earn more money on a project and minimize the developing time cost. Sometimes due to unavailability of experts may decrease the profit of the organization. Thus, the organization comes with these types of issues. They want to increase the profit and decrease the development time and on fewer budgets, hire the more expert people to achieve the organization's goal. For this purpose, they are applying the new development approach called GSD (global software development). Through global software development, they develop their project on a reasonable budget and maximize profit. However, in GSD, other challenges of team communication, coordination, geographical location, and cultural and time zone differences increase the project's effort. This paper shows the more critical factors in GSD and the more challenging and shows their impact. They are more challenging on which project manager or team leader more focus on these factors, which can be more helpful for the project's success. This paper is more helpful for both industries and researchers in that they can easily estimate the effort in the context of GSD.

CCS CONCEPTS

• **Software and its engineering** → **Distributed Software Development** ; • **Effort Estimation** → *Global Software Development*; • **Effort Estimation In Global Software Development**;

KEYWORDS

Global Software Development (GSD); Cost Drivers; Software Effort Estimation

ACM Reference Format:
Danish Iqbal, Ijaz ul haq and Irfan Kazim. 2022. The Influence of Cost Drivers on Effort Estimation in Distributed Software Development. In *The International Conference on Evaluation and Assessment in Software Engineering 2022 (EASE 2022), June 13–15, 2022, Gothenburg, Sweden.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3530019.3531331

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
EASE 2022, June 13–15, 2022, Gothenburg, Sweden
© 2022 Association for Computing Machinery.
ACM ISBN 978-1-4503-9613-4/22/06…$15.00
https://doi.org/10.1145/3530019.3531331

1 INTRODUCTION

In Global software development, multiple teams are involved in a specific software project. All teams may consist of different people with different geographical locations, communication gap, and their capability of understanding is also different in GSD. In distributed team projects, the project manager and the development team of a particular project are distributed. They find it difficult to focus on a common point of interest and how to implement the project. In GSD, distributed team members follow different standards for development, quality models, and processes for the project's development, which leads to project failure. In [17], a study was conducted in which 40 percent of failed projects related to offshore. Several challenges are discussed related to GSD in [20], [30], and [7], which relate to project the failure of projects.

In literature, a survey was conducted on the algorithmic model in which 364 companies' data were examined. Almost 14 percent of organizations used an algorithmic model for cost estimation, and other organizations used analogy base estimation [10]. Another research shows that 17 percent of organizations are using component-based development. Almost 7 percent of total organizations use algorithmic models for estimation [17]. However, the most regular and common technique which is used for cost estimation is past project base.

Some studies are conducted in which they found factors affecting contextual base communication. They evaluate that there is no proper framework to solve it. In Global Software Development (GSD) the achievement and failure of any software depends on the very significant factor, which is geographical location. In Global Software Development (GSD), requirements gathering face is most difficult task some work is done in literature [30], but they only focus on the management of requirements, and they claimed that in GSD requirement managing is the most difficult task.

There are number of reasons for algorithmic models are not used in different organizations because these models are unable to historical data or past project data. One of the previous research projects examined that 50 percent of companies [10] do not store historical data for future reuse from which they can reduce their development effort. Another research was also conducted, which determined that approximately 300 measurement programs began between 1980 and 1990, between these, 75 percent were successful [26]. By calculating performance related elements, the analogy base estimates give better results than model base estimates [21].

Historical data draw a big impact on project estimation because if our previous estimates are not accurate and consistent, then estimation for new project does not produce better results, so because

of these, chances of under and overestimation becomes high, which is not appropriate for any project [3].

In the new era of Globalization, the necessity of software and business process increases day by day, so Global Specifically, communication hurdles in GSD impose bad impact on project cost, which affect the estimation of software development effort and cost, as shown in Table 1 and Table 2. So, it is necessary to focus on communication and coordination factors, which help us estimate GSD projects effectively. GSD provides a distributed working environment for organizations to produce maximum output by making an effort to reduce code and implement innovation and ideas with scarce resources and in a limited period. Development company's trends are now changing from co-located to distribute working environment. COBRA or Cocomo 2 is used to check the impact of different geographical, social, and cultural factors involved in any project of the development organization. We identified the factors affecting GSD as portrayed in Table 3, Table 4, and Table 5 by estimating their impact on cost, project success, and risk using different estimation techniques and finding results, and concluding the paper at the end.

The rest of this article is structured as follows : In Section II, we briefly introduce the related work of GSD. In Section III, we described our methodology. In Section IV, we analyze the experimental results in detail. In Section V, we conclude our study.

2 RELATED WORK

Global Software Development (GSD) refers to a set of activities, processes, practices, and technologies that people and businesses use to create and maintain software and products [22]. In GSD Communication barriers in GSD have a negative influence on project costs, as illustrated in Tables 1 and 2. Distance (the distance between developers and their clients or end users); time-zone differences (to a significant extent, time zone is a complicating factor with distance); and national culture are the elements that distinguish GSD from normal (centralized) (including language, national traditions, customs, and norms of behavior). Distance, time zone, and national culture all have influence on the GSD, and we discovered the components that affect it from the literature. Table 3 illustrates these factors, and Table 4 shows their occurrence.

3 METHODOLOGY

In this study, the systematic literature review (SLR) methodology is used the identification different. Factors Affecting the Effort Estimation in Global Software Development (GSD). SLR is a style of study that is used to discover a specific problem to identify, analyze and explore [11].

3.1 Systematic Literature review

To identify, analyze, and explore a specific problem or topic, the systematic literature review (SLR) is the best method for identifying and evaluating the problem. SLR provides deep knowledge about a specific situation from the literature and applies the inclusion and exclusion method [31], [27] to select particular papers. In the Systematic Literature Review shown in Table 7, there are main three phases which are planning, conducting, and reporting and shown in Table 6.

3.2 Phase 1

Planning

Research questions:

This study aims to identify different factors affecting effort estimation in global software development (GSD). Ours focused on the research question, which is defined below.

RQ1: What factors affect the Effort Estimation in Global Software Development (GSD)?

Inclusion Exclusion Criteria: In this study, we have selected the papers written in the English language. In these papers, complete information must be presented and analyzed using some methods. The selected papers for this study must be published or presented in any workshop, conference, or journal. We have studied or selected those papers in which different factors affecting on global software development (GSD) while estimating the effort.

Exclusion criteria In this study, we have rejected those papers written in remaining languages except English, and the papers selected multiple times were also excluded. In papers where factors affecting effort estimation in global software development are not discussed are rejected.

Quality Assessment criteria for study selection n identifying different factors involved in GSD, we parallel achieved the Quality Assessment mechanism for the nominated papers with the information extraction phase. For QA, guidelines represented in [3] [17] [30] are followed, and a checklist is also produced.

- Value 1 is assigned to papers that fully represent all the questions present in the checklist
- Value 0.5 is assigned to papers that do not fully represent all the questions present in the checklist but partially.
- Value 0 is assigned to those papers which do not represent any question present in the checklist

Search String Search strings are based on different keywords related to a specific topic, and research strings are derived from literature and research questions [9][10][11][12]. In this paper, the following search strings, as presented in Table 7, are used for different databases for the selection of papers.

3.3 Phase 2

Conducting Primary study:

Primary studies are analyzed in this phase. The papers selected in the preliminary study are finalized using the approach tollgate proposed in [27]. Different phases of the tollgate approach are discussed below.

Phase 1. Relevant papers are selected by using the search string applied to different databases.

Phase 2. In the second phase, apply inclusion and exclusion based on the paper title.

Phase 3. This phase applies the inclusion and exclusion based on the paper abstract and conclusion..

Phase 4. In phase 4, apply inclusion and exclusion after reading the full paper text.

Phase 5. After inclusion and exclusion apply to previous phases, the remaining papers are selected for the SLR.

Data Extraction

Table 1: Questions, Concept Model, and the Findings

Reference	Focus Questions	Concept Model	Findings
[15]	Identifying different functions affecting GSD most in term of complexity and draw causal model and find most critical factors affecting the GSD.	21 most common factors were identified for GSD. Divide these factors into seven different categories and analysis were performed using structural equation.	Number of different field are identified in which work can be extended difficulties related to team and the impact of estimation in project duration.
[22]	Factors identifying that impact software team productivity and affect the schedule in global software development.	Three years of project data related to GSD are analyzed and different factors are identifying related to team productivity and model is proposed which helps to distinguish schedule risk.	Interviewed from different team members about their problems and they say to understand specific functionality in depth communication is needed and the expert person contact was unavailable.
[25]	Requirements engineers face lack of knowledge or framework for identifying different factors that affect the requirement engineering process.	In industry, survey was conducted to evaluate different factors that affect RE activities and these factors are accommodated in RE framework.	Finding the prominent factors, we calculate the mean for each factor for this purpose. The factors have value equal or above 4 and are consider as a most powerful factor.
[16]	Language, geographical and culture differences are the major problems in communication while in GSD text base and audio base communication is very important and it requires improvement according to GSD context.	In these 57 interviews were conducted on 8 GSD projects and all projects used audio and text base communication.	In this study, they divide the GSD users in two categories like technical and non-technical and identified that technical users use text base communication and non-technical users use audio base communication.
[16]	Analyzing different techniques that evaluate performance of learner in GSD and how they affect global software development.	A study was conducted in which different students assigned projects total 70 students were participated in GSD project. Students are dividing into 15 different groups.	In this research, the authors are unable to define how to analyze the groups contributing to GSD. In this study three assessment approaches were used and none of them able to address all related issues.
[31]	Offshore and onsite team draw a drastic impact in global software development, so identification of different factors affecting on these elements is important.	In this research 10 different factors were used for evaluating the commitment between offshore team and onsite team. Using this empirical knowledge, we determine that knowledge sharing between offshore and onsite team have major impact on GSD.	Finding of this paper explain that how onsite team and offshore team expertise, commitment and trust have very important effect on the outcome or work produced in GSD.
[18]	How to handle the change requirements in GSD? How to calculate the estimation and maintain when some changes occurs. How to analyze the change when comes in GSD and make a project to success.	They handle the change requirement and minimize the cost and time. Improved the chances of developing toward the success in GSD. They conduct the case study for online shopping system. They handle these changes. They validate their method on case study.	They cannot estimate the whole effort and cost and how much they reduce the cost of change from their method. They applied their method only on one application they did not apply on commercial projects.
[5]	What are the factors that directly influence on effort estimation?	They reduce the cost and time through better effort estimation. They apply different effort estimation method just as Cocomo and SLIM. They validate their method through expert judgment.	They cannot find the influence on risk. What's the impact on project success or failure they did not talk about it? But there are also some barriers comes when they validate their method through expert depend on expertise level and skills.
[23]	Which methods are used in GSD for effort estimation? What's the factor and sub factor that influence on GSD? Which type of industry they target and which type of project?	They apply some use cases and apply some estimation method and validate the results. They work on agile method.	The main barrier they note the time zone and cultural.Their results do not show us how much they minimize the cost and time and what's the ratio of the project success and failure.
[14]	Work distribution and task allocation strategies and factor that influence on them in globally distributed environment.	By conducting Systematic Literature Review and Empirical Studies Identified 11 factors related to task allocation support to improve estimation accuracy in GSD.	Need focus on different task allocation criteria, standard rules and procedures for accurate task allocation in order to make accurate estimation of software projects under GSD. For the validation of these finding and to check the impact of these factor on task allocation empirical studies were conducted.

Table 2: Identified Factors from Selected Papers

Identified form Paper [20]	**Factors identified form Paper [10]**	**Factors identified from Paper[26]**	**Factors identified from Paper[3]**	**Factors identified from Paper[21]**
Communication Cultural Travel Process Time Zone	Identifying different functions affecting GSD most in term of complexity and draw causal model and find most critical factors affecting the GSD.	21 most common factors were identified for GSD. Divide these factors into seven different categories and analysis were performed using structural equation.	Number of different field are identified in which work can be extended difficulties related to team and the impact of estimation in project duration.	Language and Cultural Differences, Time Zone, Trust, Communication, Requirements Stability, Evolution in Technology, Team Experience, Face to Face Interaction
Identified form Paper [28]	**Factors identified form Paper [22]**	**Factors identified from [25]**	**Factors identified from [16]**	**Factors identified form [11]**
Team Performance, Task Dependency, Communication, Language and Cultural Differences, Task Uncertainty	Requirements Stability, Geographical Distance, Time Zone Differences, Task Size, Team Experience, Task Dependency, Government Rules	Project Manager Skills, Requirements Specification, Communication ,language and Cultural Differences, Architecture Mismatch, difficulties related to team and the impact of estimation in project duration.	Team Experience, Face to Face Interaction, Evolution in Technology, Language and Cultural Differences, Time Zone Differences	Scarce of Communication, Scarce of Documentation, Scarce of Compatibility, Mismatches
Identified from [31]	**Factors identified from [19]**	**Factors identified from [5]**	**Factors identified from [18]**	**Factors identified from [8]**
Cultural, Communication, Lack of Integration Planning, Roles Responsibilities, Lack of Knowledge and Skills	Language Difference, Cultural Difference, Communication Gap, Team Capability	Continuous Validation, Efficient Utilization of Resources, Time Zone Differences, Communication Difference, Task Dependency, Process Maturity	Different Time Zone, Delay in Response, Unavailability of Concerned Personal, Communication, Team Experience	Requirements Stability, Coupling between Tasks, Task Complexity, Technical Knowledge, Project Experience, Language and Cultural Differences
Identified form Paper [8]	**Factors identified from [23]**	**Factors identified from [14]**	**Factors identified from [12]**	**Factors identified from [3]**
Project Manager Experience, Team Leader Behavior, Conflicts in Team, Task Dependency	Requirements Understanding, Team Trust, Architecture Mismatch, Lack of Integration Planning.	Language and Cultural, Geographical Distance, Process and Management Issues, Product Architecture, Trust in Team.	Early Identification of Dependencies, Communication Tools, Documentation, Integration, Process Maturity.	Travel, Team Size, Requirements Stability, Unrealistic, Milestones, Time Zone, Language and Cultural Differences

After applying the inclusion and exclusion criteria based on the tollgate approach, selection of papers. As shown in Figure 1. **Data analysis**

Different factors affecting global software development while estimating the effort are identified from the papers selected from the literature. We use the tollgate approach for this purpose. We also assessed the research question from the data achieved from the literature.

3.4 Phase 3

Reporting

This study aims to extract different factors affecting global software development while estimating the effort. To achieve this study,

Table 3: Finalize Factors after Removing Duplication

Communication	Language and Cultural Differences	Process Maturity
Time Zone	Evolution in Technology	Unrealistic Milestones
Time Zone	Evolution in Technology	Unrealistic Milestones
Requirements Stability	Task Dependency	Task Complexity
Team Experience	Team Size	Team Trust
Architecture Mismatch	Lack of Integration and Planning	Integration Failure
Communication with face-to-face	Unclear Responsibility	Lack of Communication
Lack of Documentation	Geographical Distance	Continuous Validation
Conflicts in Team	Requirements Understanding	Team organization

Table 4: Numbers of occurrence of different factors

Communication	**Language and Cultural Differences**	**Process Maturity**	**Process Maturity**	**Process Maturity**	**Process Maturity**
Evidence forms papers PE=paper evidence	PE[17], PE[20], PE[26], PE[21], PE[28], PE[22], PE[25], PE[16] , PE[11], PE[31], PE[27], PE[2], PE[5], PE[8], PE[14], PE [12], PE[3]	PE[20], PE[22] , PE[22], PE[16], PE[11], PE[31], PE[27], PE[18],PE[5], PE[23],	PE[20], PE[10], PE[26], PE[25], PE[16], PE[11], PE[31], PE[31], PE[18],PE[5], PE[23],	PE[25], PE[25], PE[11],PE[23]	PE[1], PE[20], PE[21], PE[26], PE[22], PE[5], PE[3]

a detailed systematic literature review is conducted. We identify factors from the literature on global software development that affects GSD while estimating the effort. In future work, we will provide authentic solutions to overcome the most common factors in GSD.

4 RESULTS

After the text edit has been completed, the paper is ready for the template. The results show that the impact of communication-related factors is very high as compared to any other factors because due to software development at the distributed location, the most critical challenges are communication challenges such as lack of face-to-face interaction, language barriers, cultural differences, geographical differences and time zone difference which will cover 25% highest as compare to all other factors. After that, the coverage of management-related factors is covered the 23% of overall factors the management related factors that highly affect the effort estimation because if the management overhead increases, the effort also increases the categorical factors included in management that highly affect the overall result of simulation the challenges are team cost, schedule and resources used by the development organizations. The results of organizational factors also impose maximum impact on effort estimation compared to process and technology-related factors. The organizational factors which covered 20% that highly affect the effort estimations are unavailability requirements, team conflict, project management experience, and lack of documentation. Technological factors have the 17% contribute to the overall impact on effort, and the factors that impact on the goal are an evolution of technology, use of tools and architectural mismatches are also challenged that may increase the overall effort in distributed software development. The process-related factors maturity, task

Table 5: Challenging factors in GSD

Factors	Equations used/Scaling	Description	References
Communication	a(a-1) ÷ 2 Where a = total no of stakeholders participate in projects.	If communication channel increase the more effort required.	Software Project Management Book sixth Edition
Management	SPI = EV/PV, SCI = EV/AC If SP! >1 behind schedule, If SP! <1 ahead schedule, If SP!=1 exactly on schedule.	Poor project management may increase project effort	Software Project Management Book sixth Edition
Process	ROE = net income/ average stakeholder equity Total ROE = ROE1 + ROE2 + ... + ROEn (-infinity to 0%) poor (0% to 2%) normal (2% to 4%) good (4% to + infinity) v.good	If process is mature less effort required. [equity= total assets-total liabilities) 0-1 CMMI Level poor 2-3 CMMI Level good 4-5 CMMI Level very good	[11]
Organization resources	New project resources * training overhead / 100	Experience resources required to train new resources so effort increase.	[29]
Project Complexity	Project size + time/ no of requirement * 100	If project is complex the chance to error increase so effort also increase	[4]
Team Conflict	1 high conflict (46 %), 2 medium (26 %), 3 low conflict (17 %-), 4 No conflict (12 %-)	1. due to organizational law imposed on customer 2. due to misunderstanding org laws or policies 3. don't accept law conflict	[13]
Tool used	Total tool = communication tool + cooperation tool + control tool	If no of tool increase, then in will increase effort due to training overhead of new tool	[6]
Architectural change	((n/total)*100), n= missing requirement, n= add requirement + delete requirement + modify requirement	If architectural requirement changes the effort increased	[24]
Programmer Productivity	Productivity = LoC/PM or FP/PM	Programmer productivity is increase then effort also increase.If the accuracy of programmer productivity increase the effort reduced.	[9]
Requirement understanding	No of total requirements, unclear requirements / total requirements	When requirement are unclear then its difficult to understand so effort increase	[13]

dependency, and continuous validation have overall combine 15% effect on effort estimation.

5 CONCLUSIONS

This paper shows the result of a systematic literature review of effort estimation in the environment of GSD. The paper's main goal is to improve productivity and minimize the cost of the effort estimation in GSD. It is helpful for both industry and researcher. We have identified the 26 factors that affect effort estimation and categorized them into five main categories. The result shows which factor is more challenging and critical regarding effort estimation in GSD. From these factors, we conclude that we simulate the factor

Table 6: Phases of SLR

Phases of SLR	Steps
Planning	**1**.Select Databases to get the Papers 2. Identify Research Questions 3. Identify and apply Inclusion and Exclusion Criteria 4.Identify the Search String
Planning	**1**.Select Databases to get the Papers 2. Identify Research Questions 3. Identify and apply Inclusion and Exclusion Criteria 4.Identify the Search String
Conducting	**1**.Conduct Primary study (Background Knowledge) 2. Data Extracted from Different Databases Analyze the Data
Reporting	**R**esults are Represented and well Documented

Table 7: Data gathering source

Databases	Search String
IEEE Explorer	**(**Factors Or variables) AND (Effort Estimation) AND (“Global Software Development” OR “Global Software Engineering”)
ACM	**(**Factors Or variables) AND (Effort Estimation) AND (“Global Software Development” OR “Global Software Engineering”)
Wiley Inter Science	**(**Factors Or variables) AND (Effort Estimation) AND (“Global Software Development” OR “Global Software Engineering”)
Google Scholar	**(**Factors Or variables) AND (Effort Estimation) AND (“Global Software Development” OR “Global Software Engineering”)
Science Direct	**(**Factors Or variables) AND (Effort Estimation) AND (“Global Software Development” OR “Global Software Engineering”)

Table 8: Data gathering source

E-Databases	Total	Title Filtration	Abstract Filtration	Complete Text Filtration	Selected Articles
IEEE Explorer	14	11	**7**	**6**	**5**
ACM	**8**	**6**	**5**	**3**	**1**
Wiley online	78	18	**8**	**6**	**4**
Google scholar	305	14	**9**	**8**	**7**
Science direct	87	20	11	**7**	**3**
Total	492	69	40	30	20

through simulation and validate the equation of each factor. The results show that the communication category is more challenging in a global software environment. The project manager must focus on the communication barrier because of communication, and we fail our project in the GSD environment. This SLR finds out a wide range of cost drivers that affect the project communication, cultural time zone, etc.so keep in mind when you estimate the effort in GSD. It is a very helpful to study for that industry who are working in a GSD environment through this study, we can easily estimate the effort in a distributed environment and minimize the budget of the project. In a future we want to map these factors with cocomo2 and we easily estimate the whole cost of the project through effort in GSD.

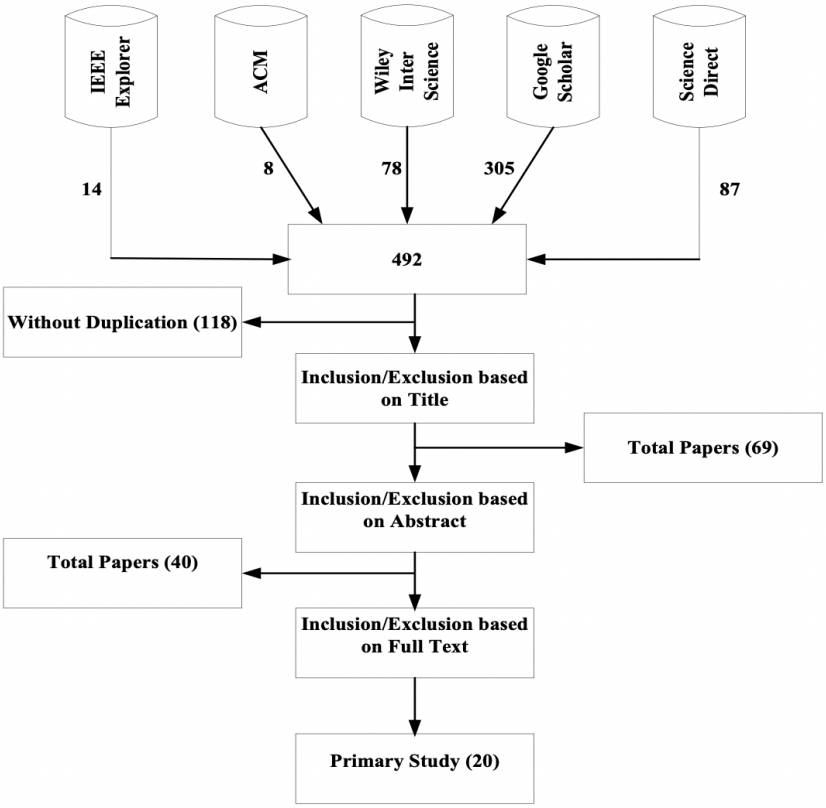

Figure 1: Article selection using tollgate approach

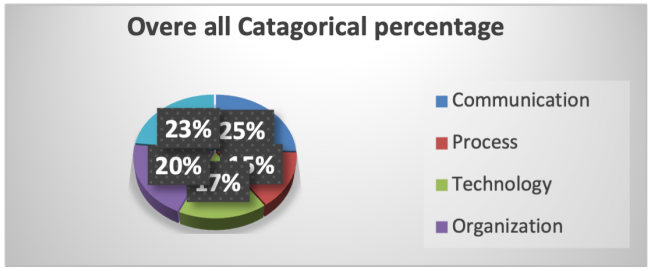

Figure 2: Overall categorical percentage

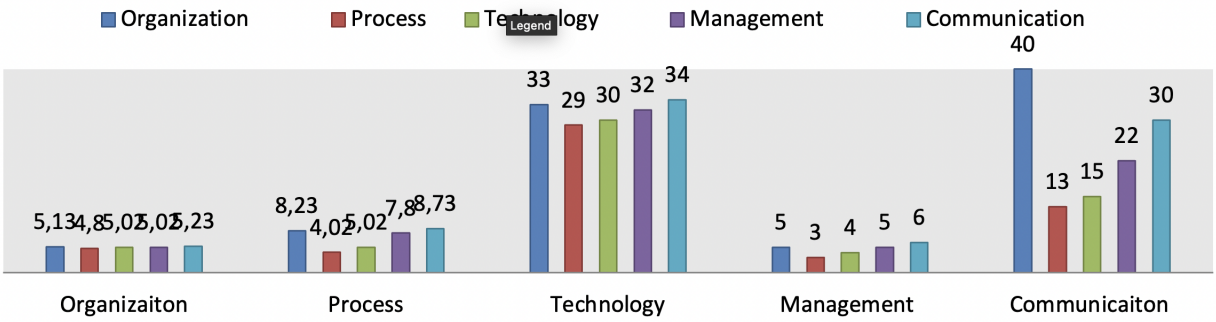

Figure 3: Graph of communication categories

REFERENCES

[1] Turki Alelyani, Ke Mao, and Ye Yang. 2017. Context-centric pricing: early pricing models for software crowdsourcing tasks. In *Proceedings of the 13th International Conference on Predictive Models and Data Analytics in Software Engineering*. 63–72.

[2] Naveed Ali and Richard Lai. 2016. A method of requirements change management for global software development. *Information and Software Technology* 70 (2016), 49–67.

[3] Julian M Bass, Beecham Sarah, Mohammed Abdur Razzak, and John Noll. 2018. Employee retention and turnover in global software development: Comparing in-house offshoring and offshore outsourcing. In *2018 IEEE/ACM 13th International Conference on Global Software Engineering (ICGSE)*. IEEE, 77–86.

[4] Nicolas Bettenburg, Meiyappan Nagappan, and Ahmed E Hassan. 2012. Think locally, act globally: Improving defect and effort prediction models. In *2012 9th IEEE Working Conference on Mining Software Repositories (MSR)*. IEEE, 60–69.

[5] Ricardo Britto, Muhammad Usman, and Emilia Mendes. 2014. Effort estimation in agile global software development context. In *International Conference on Agile Software Development*. Springer, 182–192.

[6] Saad Yasser Chadli, Ali Idri, Joaquín Nicolás Ros, José Luis Fernández-Alemán, Juan M Carrillo de Gea, and Ambrosio Toval. 2016. Software project management tools in global software development: a systematic mapping study. *SpringerPlus* 5, 1 (2016), 1–38.

[7] Mario Coccia. 2018. Measurement of the evolution of technology: A new perspective. *arXiv preprint arXiv:1803.08698* (2018).

[8] Ana Cristina Costa and Neil Anderson. 2011. Measuring trust in teams: Development and validation of a multifaceted measure of formative and reflective indicators of team trust. *European Journal of Work and Organizational Psychology* 20, 1 (2011), 119–154.

[9] Ana Cristina Costa and Neil Anderson. 2011. Measuring trust in teams: Development and validation of a multifaceted measure of formative and reflective indicators of team trust. *European Journal of Work and Organizational Psychology* 20, 1 (2011), 119–154.

[10] Fabio QB da Silva, Catarina Costa, A Cesar C Franca, and Rafael Prikladinicki. 2010. Challenges and solutions in distributed software development project management: A systematic literature review. In *2010 5th IEEE International Conference on Global Software Engineering*. IEEE, 87–96.

[11] Jose Javier Dolado. 2000. A validation of the component-based method for software size estimation. *IEEE Transactions on Software Engineering* 26, 10 (2000), 1006–1021.

[12] D Hartama, Herman Mawengkang, M Zarlis, and RW Sembiring. 2017. Smart City: Utilization of IT resources to encounter natural disaster. In *Journal of Physics: Conference Series*, Vol. 890. IOP Publishing, 012076.

[13] Danish Iqbal, Assad Abbas, Mazhar Ali, Muhammad Usman Shahid Khan, and Raheel Nawaz. 2020. Requirement validation for embedded systems in automotive industry through modeling. *IEEE Access* 8 (2020), 8697–8719.

[14] Dehua Ju. 2006. A concerted effort towards flourishing global software development. In *Proceedings of the 2006 international workshop on Global software development for the practitioner*. 62–65.

[15] Huma Hayat Khan, Mohd Naz'ri bin Mahrin, and Suriayati bt Chuprat. 2014. Situational requirement engineering framework for global software development. In *2014 International Conference on Computer, Communications, and Control Technology (I4CT)*. IEEE, 224–229.

[16] S Arun Kumar and Arun Kumar Thangavelu. 2013. Factors affecting the outcome of Global Software Development projects: An empirical study. In *2013 International Conference on Computer Communication and Informatics*. IEEE, 1–10.

[17] Adailton Magalhães Lima. 2010. Risk assessment on distributed software projects. In *2010 ACM/IEEE 32nd International Conference on Software Engineering*, Vol. 2. IEEE, 349–350.

[18] Sajjad Mahmood, Sajid Anwer, Mahmood Niazi, Mohammad Alshayeb, and Ita Richardson. 2017. Key factors that influence task allocation in global software development. *Information and Software Technology* 91 (2017), 102–122.

[19] Prateeti Mohapatra, Petra Björndal, and Karen Smiley. 2010. Causal analysis of factors governing collaboration in global software development teams. In *2010 5th IEEE International Conference on Global Software Engineering*. IEEE, 128–132.

[20] Mohammad Muhairat, Saleh Aldaajeh, and Rafa E Al-Qutaish. 2010. The impact of global software development factors on effort estimation methods. *European Journal of Scientific Research* 46, 2 (2010), 221–232.

[21] Mahmood Niazi, Sajjad Mahmood, Mohammad Alshayeb, Abdul Majid Qureshi, Kanaan Faisal, and Narciso Cerpa. 2016. Toward successful project management in global software development. *International Journal of Project Management* 34, 8 (2016), 1553–1567.

[22] Tuomas Niinimaki, Arttu Piri, and Casper Lassenius. 2009. Factors affecting audio and text-based communication media choice in global software development projects. In *2009 Fourth IEEE International Conference on Global Software Engineering*. IEEE, 153–162.

[23] S Ramacharan and K Venu Gopala Rao. 2016. Software effort estimation of GSD projects using calibrated parametric estimation models. In *Proceedings of the Second International Conference on Information and Communication Technology for Competitive Strategies*. 1–8.

[24] Md Rounok Salehin. 2013. Missing requirements information and its impact on software architectures: A case study. (2013).

[25] Kathleen Swigger, Fatma Cemile Serce, Ferda Nur Alpaslan, Robert Brazile, George Dafoulas, and Victor Lopez. 2009. A comparison of team performance measures for global software development student teams. In *2009 Fourth IEEE International Conference on Global Software Engineering*. IEEE, 267–274.

[26] June M Verner, O Pearl Brereton, Barbara A Kitchenham, Mahmood Turner, and Mahmood Niazi. 2014. Risks and risk mitigation in global software development: A tertiary study. *Information and Software Technology* 56, 1 (2014), 54–78.

[27] Aurora Vizcaíno, Félix García, José Carlos Villar, Mario Piattini, and Javier Portillo. 2013. Applying Q-methodology to analyse the success factors in GSD. *Information and Software Technology* 55, 7 (2013), 1200–1211.

[28] Minghui Wu and Hui Yan. 2009. Simulation in Software Engineering with System Dynamics: A Case Study. *J. Softw.* 4, 10 (2009), 1127–1135.

[29] Minghui Wu and Hui Yan. 2009. Simulation in Software Engineering with System Dynamics: A Case Study. *J. Softw.* 4, 10 (2009), 1127–1135.

[30] Ye Yang, Mei He, Mingshu Li, Qing Wang, and Barry Boehm. 2008. Phase distribution of software development effort. In *Proceedings of the Second ACM-IEEE international symposium on Empirical software engineering and measurement*. 61–69.

[31] Atique Ahmad Zafar, Shahela Saif, Muzafar Khan, Javed Iqbal, Adnan Akhunzada, Abdul Wadood, Ahmad Al-Mogren, and Atif Alamri. 2017. Taxonomy of factors causing integration failure during global software development. *IEEE Access* 6 (2017), 22228–22239.